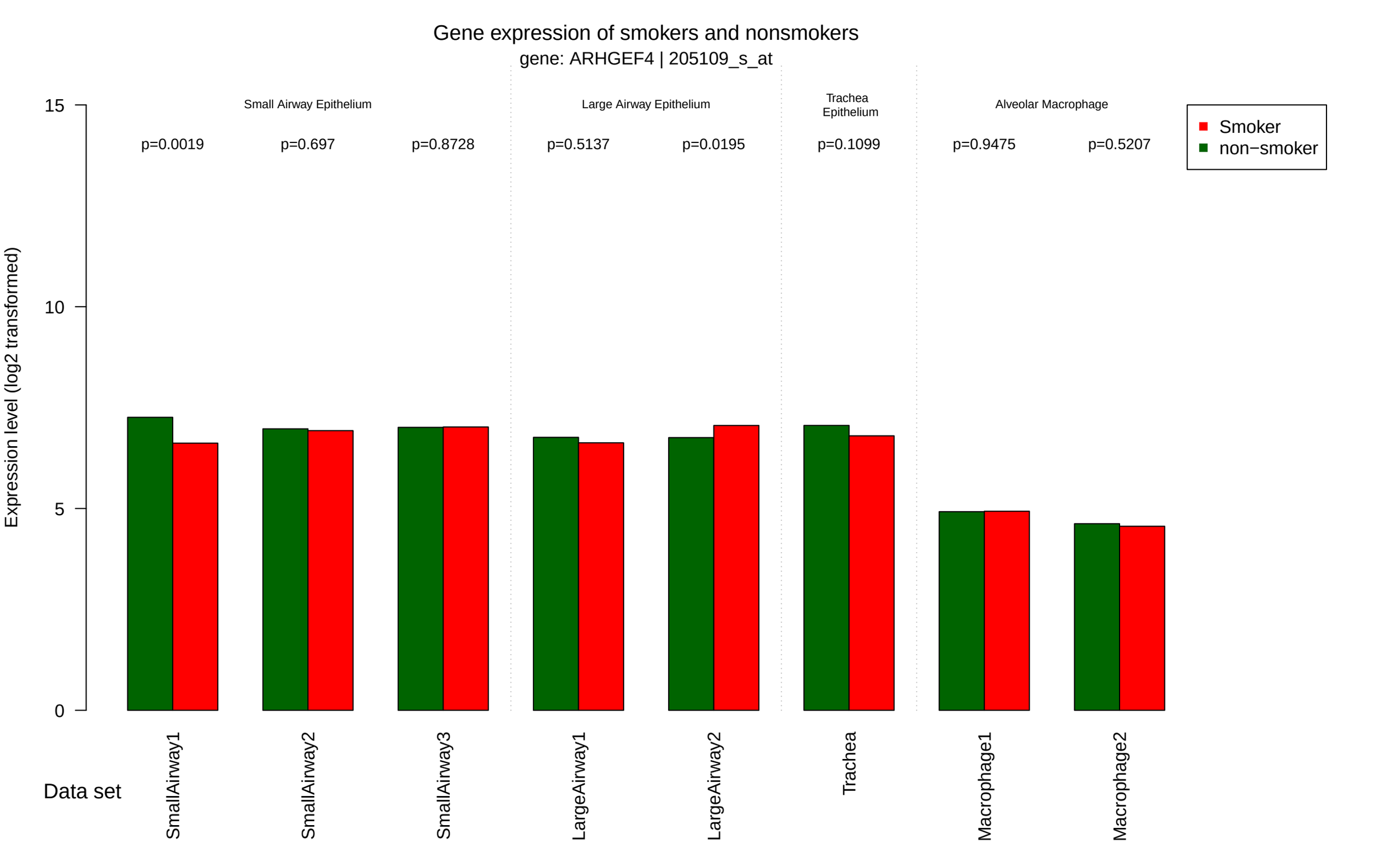
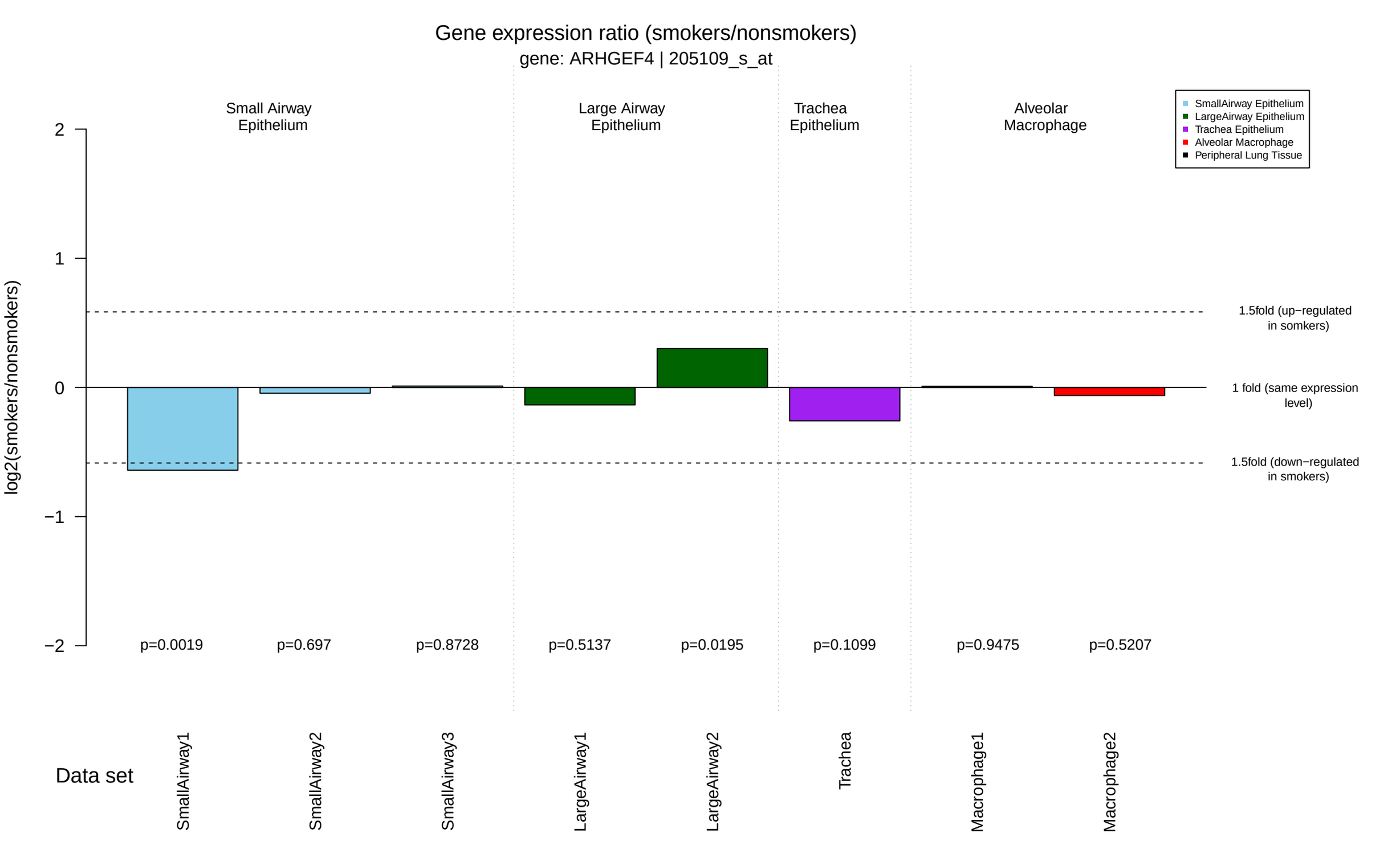
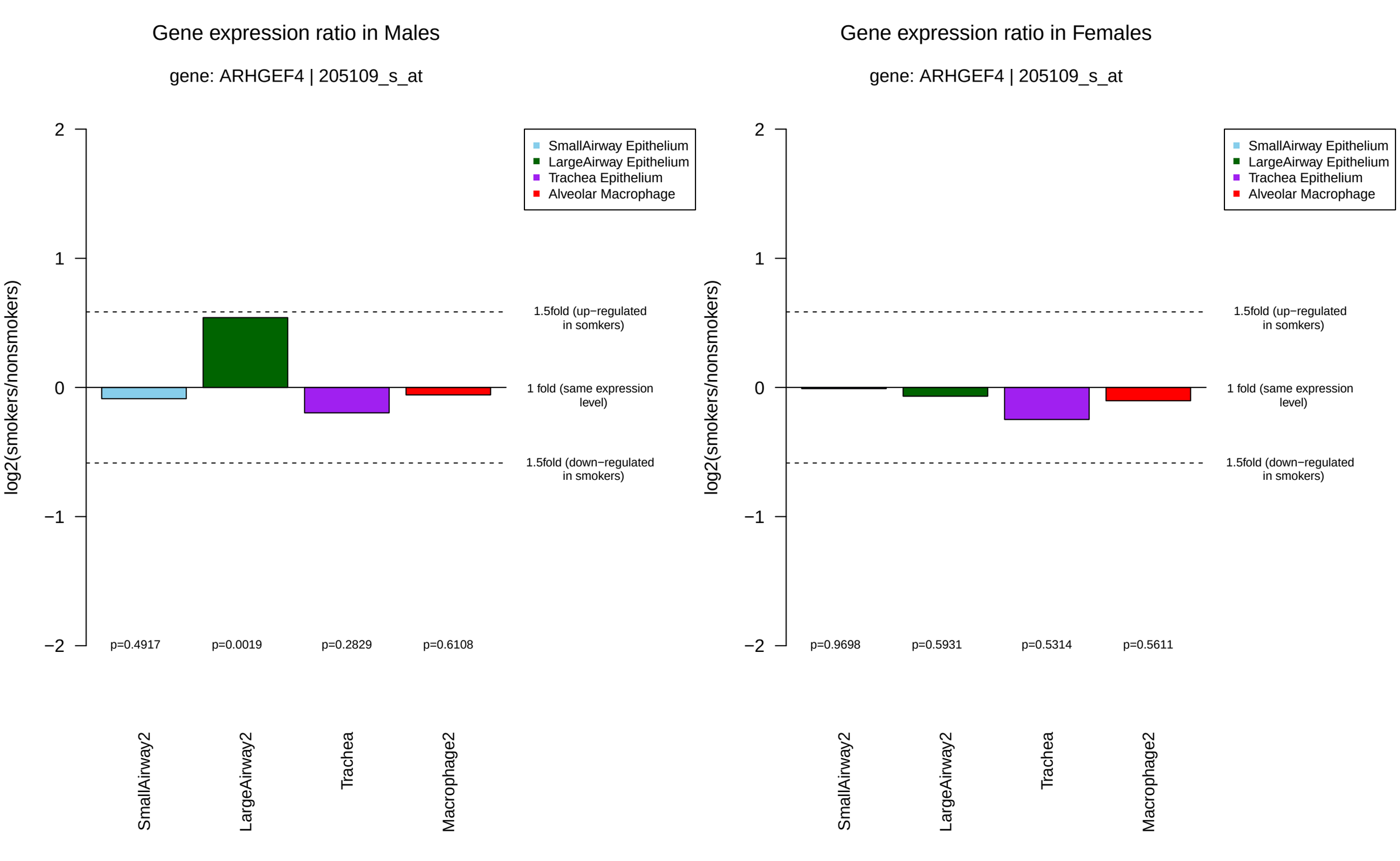
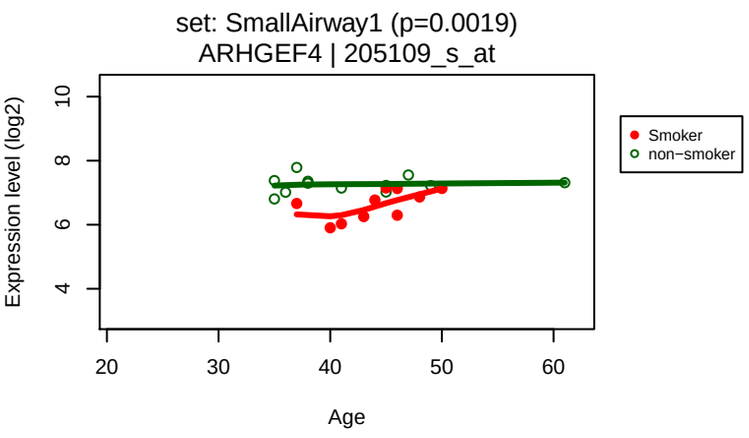
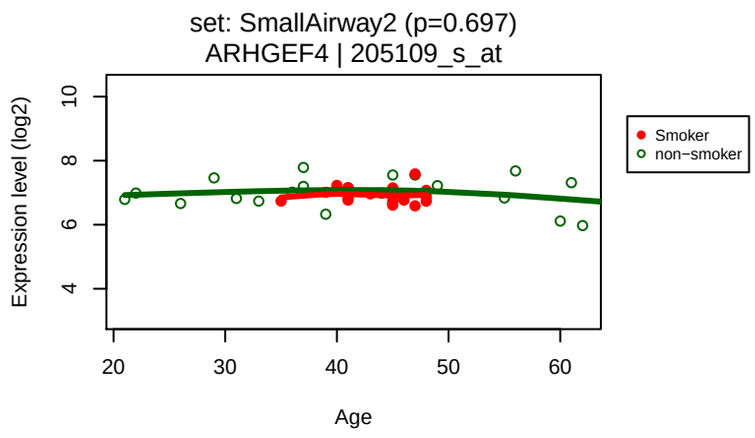
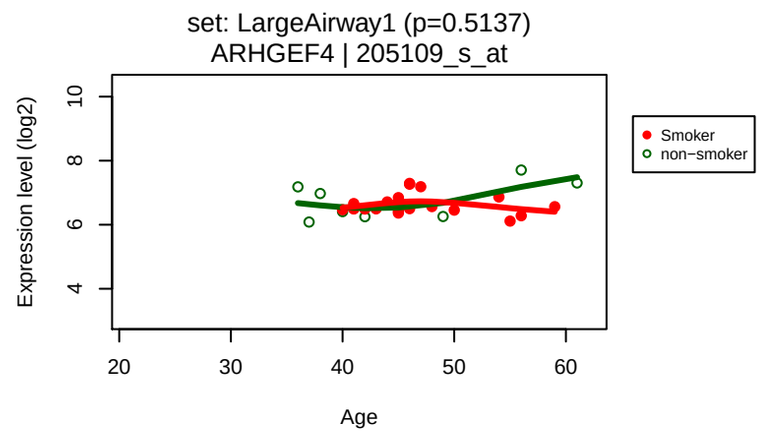

Gene expression of smokers and nonsmokers
gene: ARHGEF4 | 205109_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0019
p=0.697
p=0.8728
p=0.5137
p=0.0195
p=0.1099
p=0.9475
p=0.5207
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

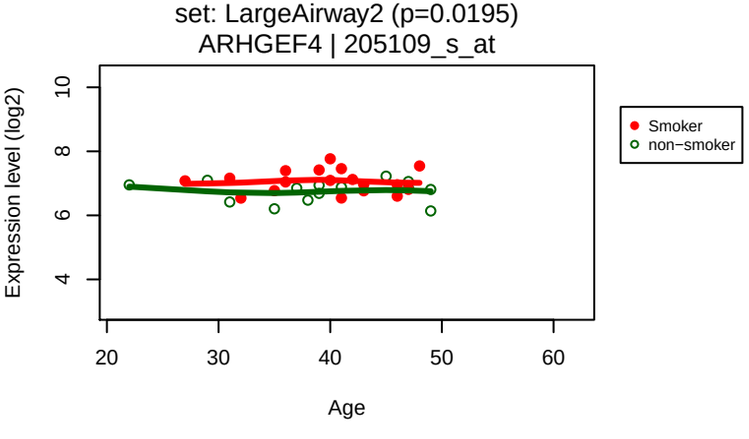

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF4 | 205109_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0019
p=0.697
p=0.8728
p=0.5137
p=0.0195
p=0.1099
p=0.9475
p=0.5207
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

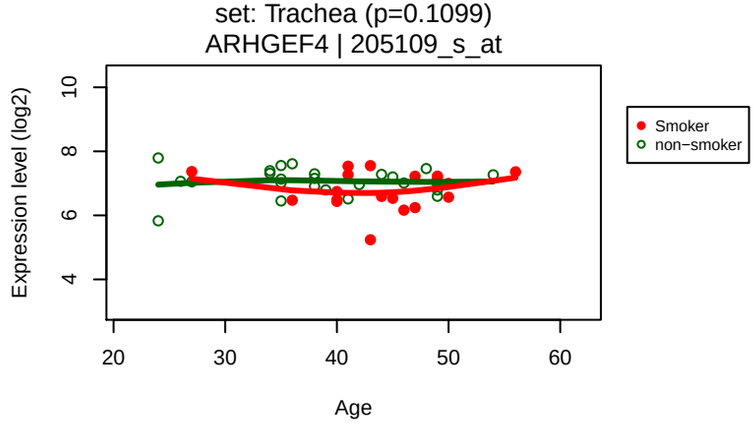

Gene expression ratio in Males
gene: ARHGEF4 | 205109_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4917
p=0.0019
p=0.2829
p=0.6108
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF4 | 205109_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9698
p=0.5931
p=0.5314
p=0.5611
SmallAirway2
LargeAirway2
Trachea
Macrophage2

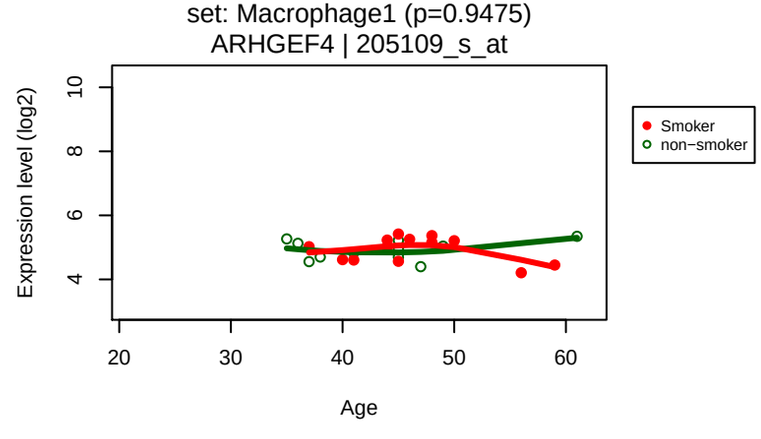

set: SmallAirway1 (p=0.0019)
ARHGEF4 | 205109_s_at
Expression level (log2)
Age
Smoker
non−smoker

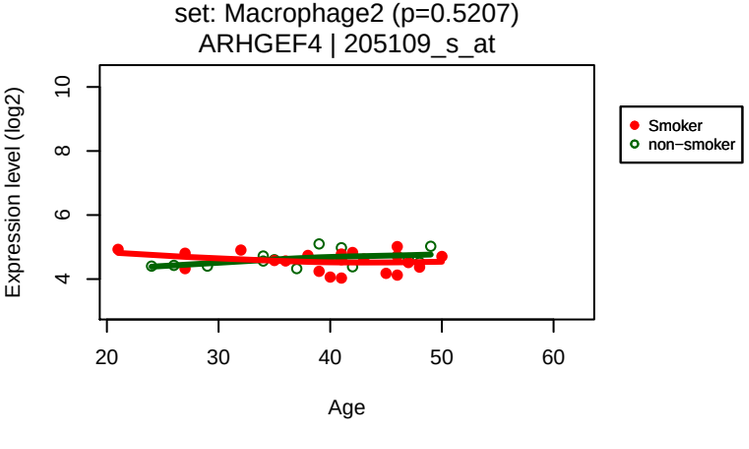

set: SmallAirway2 (p=0.697)
ARHGEF4 | 205109_s_at
Expression level (log2)
Age
Smoker
non−smoker

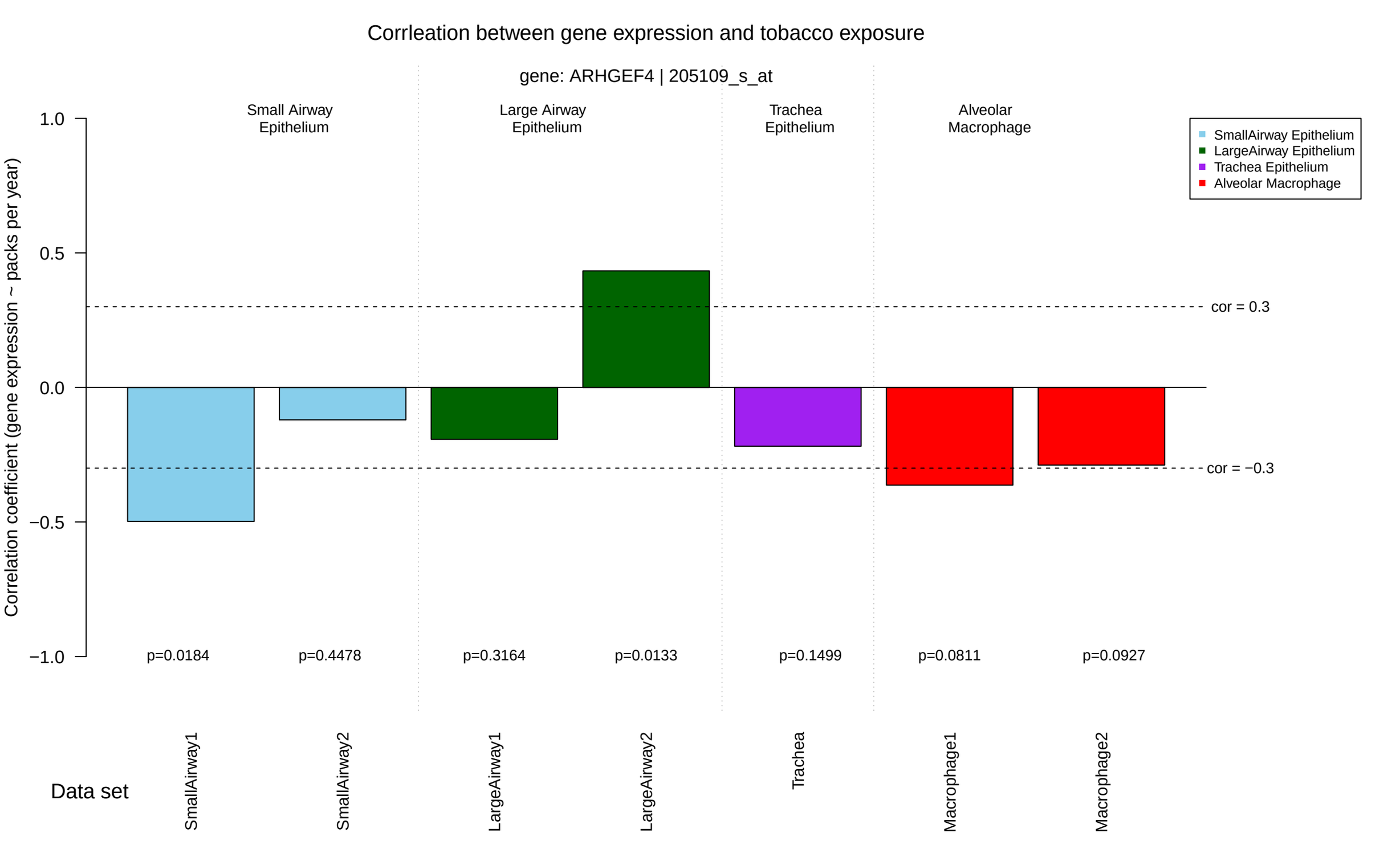

set: LargeAirway1 (p=0.5137)
ARHGEF4 | 205109_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0195)
ARHGEF4 | 205109_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1099)
ARHGEF4 | 205109_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9475)
ARHGEF4 | 205109_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5207)
ARHGEF4 | 205109_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF4 | 205109_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0184
p=0.4478
p=0.3164
p=0.0133
p=0.1499
p=0.0811
p=0.0927
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2